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We study the motion of a uniform thin rod released from rest, with the bottom end initially in contact with a horizontal surface. Our focus here is the motion of the bottom end as the rod falls. For small angles of release with respect to the horizontal, the rod falls without the bottom end slipping. For larger angles, the slipping direction depends on the static friction coefficient between the rod and the horizontal surface. Small friction coefficients cause the end to slip initially in one direction and then in the other, while for high coefficients, the end slips in one direction only. For intermediate values, depending on the angle of release, both situations can occur. We find the initial slipping direction to depend on a relation between the angle at which the rod slips, and a critical angle at which the frictional force vanishes. Comparison between experimental data and numerical simulations shows good agreement.
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I. INTRODUCTION

The plane motion of rigid bodies is an essential topic in introductory physics courses for scientists and engineers. The motion of rods and other elongated objects falling on horizontal surfaces, with one end initially in contact with the surface, is a common example used in many engineering textbooks to illustrate this topic.[1–3] However, most of the problems presented in these textbooks tend to restrict the motion to specific conditions. For example, a typical problem involves finding the angular acceleration of the object immediately after its release. Such problems, although useful, constitute only a small part of the pedagogical richness offered by such a system. For instance, straightforward and interesting questions, such as at what angle or in which direction will the contact end slip, are generally not addressed.

The fall of rods and other elongated object such as pencils, released with one end initially on a horizontal rough surface, have been studied theoretically by Turner and Pratt[4] and by Cross.[5] From the relevant equations of motion, these authors show that the end initially in contact with the surface may slide either backward ($-\hat{x}$) or forward ($+\hat{x}$), depending on the initial angle of inclination θ_0 and the coefficient of static friction μ_s. For example, if the rod is released from a stationary vertical position ($\theta_0 = 90°$) and $\mu_s < 0.371$, the rod slides backward for an initial angle greater than about 54.9°, while for $\mu_s > 0.371$ the rod slides forward for an initial angle between about 38.8° and 19.5°. Moreover, if $\mu_s < 0.371$, the rod initially slides backward for a period of time and then slides forward.

In the present paper, we study the fall of a rod from an experimental point of view. For this purpose, videos of experiments performed with a thin aluminium rod released on different surface are recorded with a high-speed camera and analyzed using a video analysis program. We find reasonably good agreement between experimental data and simulations obtained from the equation of motion.

II. THEORETICAL BACKGROUND

Figure 1 shows the free-body diagram of a uniform rigid thin rod of mass m and length L released from rest at an angle θ_0 ($0° < \theta_0 < 90°$). Let point G be the rod's center-of-mass, point O be the contact point between the rod and the flat surface, hereafter designated as the "bottom end" of the rod, and θ the angle between the rod and the horizontal surface at some time t. For a uniform density rod, the distance between points G and O is $L/2$, and the moment of inertia of the rod about point O is $\frac{1}{3}mL^2$.

From the relevant equation of motions for the rod,[4,5] it can be shown that for non-slipping conditions the angular acceleration α of the rod is given by

$$\alpha = \frac{3g}{2L}\cos\theta, \tag{1}$$

where α is considered positive if clockwise. Similarly, the frictional force f and the normal force N are given by

$$f = \frac{mg}{4}(9\cos\theta\sin\theta - 6\sin\theta_0\cos\theta), \tag{2}$$

and

$$N = \frac{mg}{4}\left(1 + 9\sin^2\theta - 6\sin\theta_0\sin\theta\right). \tag{3}$$

Equations (2) and (3) show that $f = 0$ at the critical angle $\theta_c = \sin^{-1}(\frac{2}{3}\sin\theta_0)$, while it is negative for $\theta < \theta_c$ and

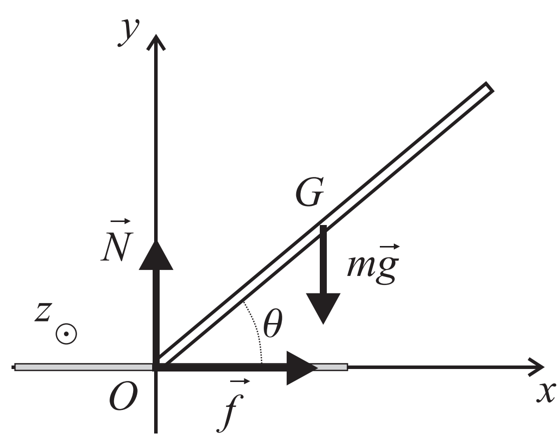

Fig. 1. Free body diagram of a rod of mass m falling in a horizontal surface.

positive for $\theta > \theta_c$. The normal force is always positive except when $\theta_0 = 90°$ and $\theta = \sin^{-1}\left(\frac{1}{3}\right)$ when it is zero. In this particular case, the rod must slip before it reaches $\theta = \sin^{-1}\left(\frac{1}{3}\right)$ no matter how large the static friction coefficient is, because $|f/N|$ becomes arbitrarily large as this angle is approached. If slipping occurs at angle $\theta = \theta_s$, then the static friction coefficient can be obtained from the condition of maximum static friction and the definition that $\mu_s = f(\theta_s)/N(\theta_s)$, giving

$$\mu_s = \frac{|9\cos\theta_s \sin\theta_s - 6\sin\theta_0\cos\theta_s|}{1 + 9\sin^2\theta_s - 6\sin\theta_0\sin\theta_s}. \qquad (4)$$

Equations (1)–(3) are valid only as long as the static frictional force is sufficiently strong to prevent the rod from slipping. If the rod slips, Eq. (1) must be replaced by[4,5]

$$\alpha = \frac{2g/L - \omega^2\sin\theta}{\cos\theta + (3\cos\theta - 3\mu_k\sin\theta)^{-1}}, \qquad (5)$$

where ω is the angular velocity of the rod and μ_k is the kinetic friction coefficient, considered positive if the bottom end is moving in the negative direction, and negative if the bottom end is moving in the positive direction. In addition, the bottom end of the rod is no longer at rest and moves with an acceleration a_O given by

$$a_O = \frac{\omega^2 L}{2}\cos\theta - \frac{\alpha L}{2}(\sin\theta - b\mu_k), \qquad (6)$$

where $b = (3\cos\theta - 3\mu_k\sin\theta)^{-1}$.

III. EXPERIMENTS AND VIDEO ANALYSIS

In this section, we report on experiments performed with an aluminium cylindrical rod of mass $m = 55.4$ g, length $L = 25.4$ cm, and radius $R = 1.5$ mm. The rod was released with the bottom end in contact with the surface of a steel sheet, a marble stone plate, and a mouse pad. A bubble level was used to make sure the surfaces were horizontal. All experiments were recorded with a high-speed video camera operating at 500 frames per second and analyzed using the video-analysis program *Tracker*.[6] Prior to analysis, the videos were spatially calibrated using a ruler (present in the videos), and an origin assigned to the initial position of the bottom end of the rod. Using *Tracker*, both the angle θ between the rod and the horizontal surface and the horizontal position of the bottom end of the rod (x_O) were measured as a function of time for different angles of release. Before the main experiments, the static friction coefficient between the rod and each of the flat surfaces was calculated using Eq. (4). For this purpose, the slipping angle was determined from videos of the rod released from a nearly vertical position. Results are shown in Table I. Note that the experimental error associated with some of these values is quite large because of the difficulty in determining the exact moment of slipping from the videos. Finally, three angles of release in the range 70°–80°, 20°–30°, and 5°–10° were selected in order to achieve $\theta_0 > \theta_s$, $\theta_0 < \theta_s$, and $\theta_0 \ll \theta_s$ for all surfaces.

A. Rod released on a steel surface

Typical video snapshots of the rod falling on a steel surface are displayed in Fig. 2 for $\theta_0 = 75.8°$. Figure 3 shows the position of the bottom end of the rod as a function of the falling angle θ for initial angles of 75.8°, 26.0°, and 5.6°. For $\theta_0 = 75.8°$, three regimes can be clearly distinguished. From the moment of release [Fig. 2(a)] until $\theta = 70.0°$ [Fig. 2(b)], the bottom end remains at rest, hence the rod falls without slipping. Beginning at $\theta = 70.0°$ the rod begins to slip backward and the bottom end moves to the left. At $\theta = 19.5°$ [Fig. 2(c)] the bottom end momentarily comes to rest as it changes direction. The rod then slips forward as the bottom end moves to the right until the end of the fall [Fig. 2(d)]. When the rod is released from $\theta_0 = 26.0°$, a similar behavior is observed as the bottom end moves first in the negative direction and then in the positive direction. However, in this case the rod begins to slip the moment it is released. Finally, for $\theta_0 = 5.6°$, there is no slipping as the bottom end remains at rest during the entire fall.

Table I. Angle of release θ_0, slipping angle θ_s, and corresponding friction coefficient μ_s.

Surface	θ_0 (deg)	θ_s (deg)	μ_s
Cloth	85.0 ± 0.2	33.5 ± 1.0	1.9 ± 0.3
Marble	85.5 ± 0.2	36.2 ± 1.0	0.9 ± 0.3
Steel	86.0 ± 0.2	67.6 ± 0.5	0.27 ± 0.01

B. Rod released on the cloth surface of a mouse pad

Figure 4 shows plots of the bottom end position when the rod is released on the cloth surface of a common computer mouse pad. For $\theta_0 = 9.8°$, the bottom end remains at rest during the entire fall and hence the rod falls without slipping. On the other hand, for $\theta_0 = 26.6°$ and $\theta_0 = 74.2°$ the bottom end remains at rest for a while and then moves to the right (video snapshots for the rod released from $\theta_0 = 74.2°$ are displayed in Fig. 5). Thus, the rod first falls without slipping and then slips forward for these initial angles. The slipping

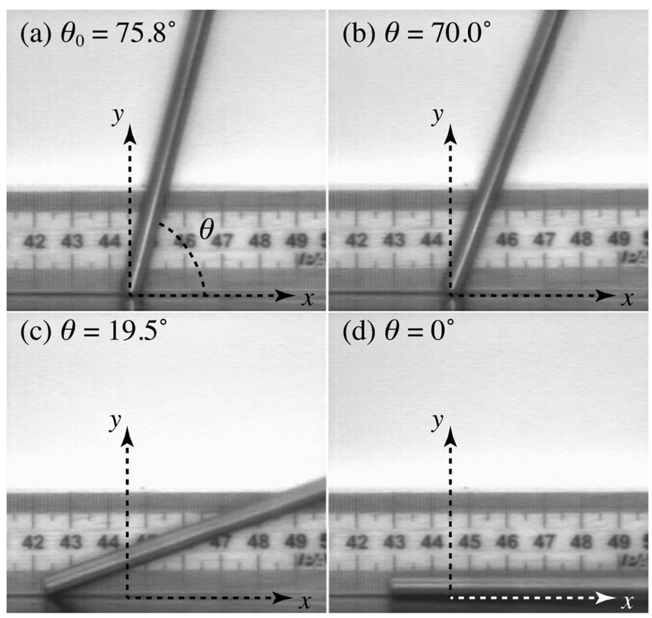

Fig. 2. Video snapshots of a rod released on a steel surface from an initial angle of 75.8° (enhanced online) [URL: http://dx.doi.org/10.1119/1.4934950.1].

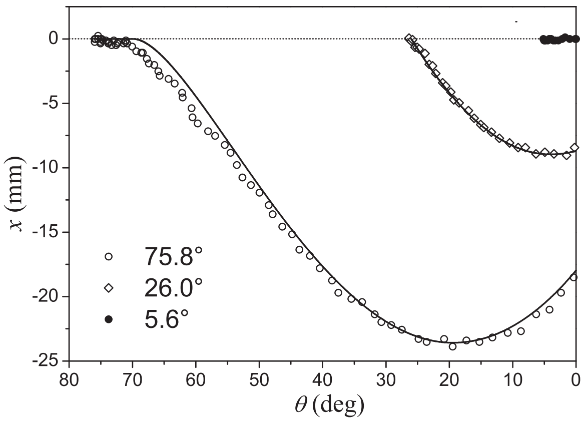

Fig. 3. Horizontal displacement of the bottom end of the rod as a function of θ when the rod is released on a steel sheet from initial angles of 75.8°, 26.0°, and 5.6°. The solid lines are numerical solutions obtained from the equations of motion.

angles were estimated to be $\theta_s = 31° \pm 1°$ for $\theta_0 = 74.2°$ and $\theta_s = 9 \pm 2°$ for $\theta_0 = 26.6°$.

C. Rod released on a marble stone surface

Figure 6 shows plots of the position of the bottom end of the rod when the rod is released on a marble stone surface. Similar to when released on a mouse pad, when the rod is released from $\theta_0 = 77.0°$, the bottom end remains at rest initially and then moves in the positive direction. However, for $\theta_0 = 27.1°$, a completely different behavior is observed. Here, the bottom end first moves in the negative direction and then moves in the positive direction near the end of the fall. This behavior is similar to the rod falling on the steel surface. For these cases, the slipping angles were estimated to be $\theta_s = 35° \pm 1°$ for $\theta_0 = 77.0°$ and $\theta_s = 26.5° \pm 0.3°$ for $\theta_0 = 27.1°$. Finally, when the rod is released from $\theta_0 = 6.4°$, the bottom end does not move at all and the rod falls without slipping.

IV. COMPARISON TO THEORY

The initial slipping behavior of the rod can be predicted theoretically by comparing θ_c and θ_s: for $\theta_s < \theta_c$ the frictional force is negative at the moment of slipping and hence the rod slips forward; for $\theta_s > \theta_c$ the frictional force is

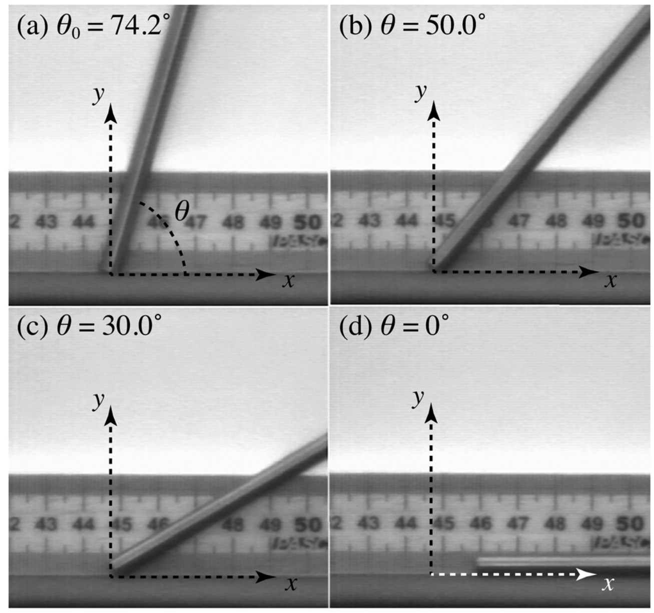

Fig. 5. Video snapshots of the rod falling on the cloth surface of a mouse pad released from an initial angle of 74.2° (enhanced online) [URL: http://dx.doi.org/10.1119/1.4934950.2].

positive and the rod slips backward. However, because θ_c and θ_s are related to θ_0 and μ_s via $\theta_c = \sin^{-1}(\frac{2}{3}\sin\theta_0)$ and Eq. (4), the initial slipping behavior can also be predicted from μ_s and θ_0. Figure 7 shows a phase diagram in the μ_s–θ_0 plane that shows the different behaviors that can occur. For $\mu_s > 0.75$ we always have $\theta_s < \theta_c$ and the rod does not slip backward. On the other hand, for $\mu_s < 0.75$ we may have $\theta_s < \theta_c$ or $\theta_s > \theta_c$ depending on θ_0, and hence the rod may slip in either direction (or remain still). Notice that slipping can always be prevented by making $\theta_s \leq 0$, which can be accomplished by making μ_s very large or by making θ_0 very small.

In order to compare the experimental results with theoretical predictions, the horizontal position of the bottom end of the rod was determined by numerically solving the equations of motion of the rod. For this purpose, the angular acceleration is first determined using Eq. (1) if falling is without slipping, or Eq. (5) if falling and slipping. The (new) values of ω

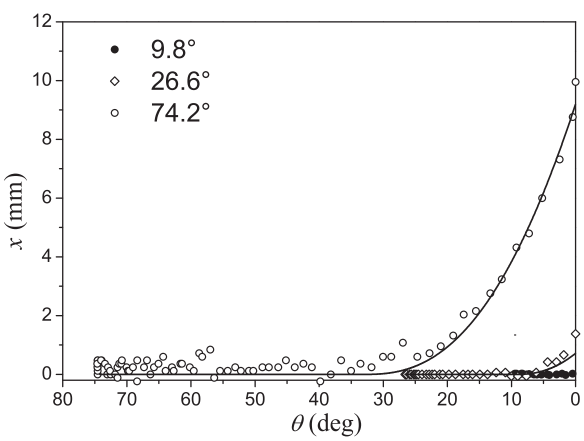

Fig. 4. Horizontal displacement of the bottom end of the rod as a function of θ when the rod is released on the cloth surface of a computer mouse pad from initial angles of 74.2°, 26.6°, and 9.8°. The solid lines are numerical solutions obtained from the equations of motion.

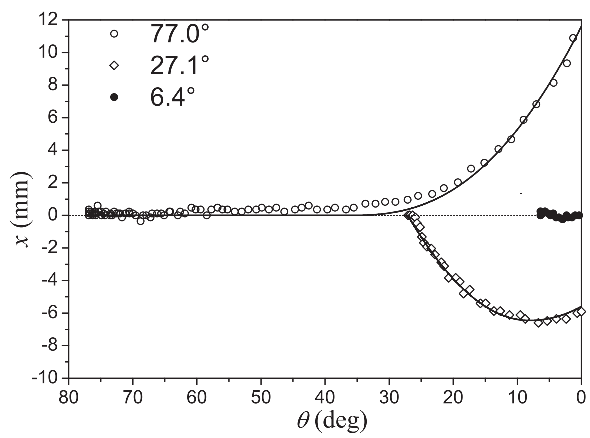

Fig. 6. Horizontal displacement of the bottom end of the rod as a function of θ when the rod is released on a marble stone from initial angles of 6.4°, 27.1°, and 77.0°. The solid lines are numerical solutions obtained from the equations of motion.

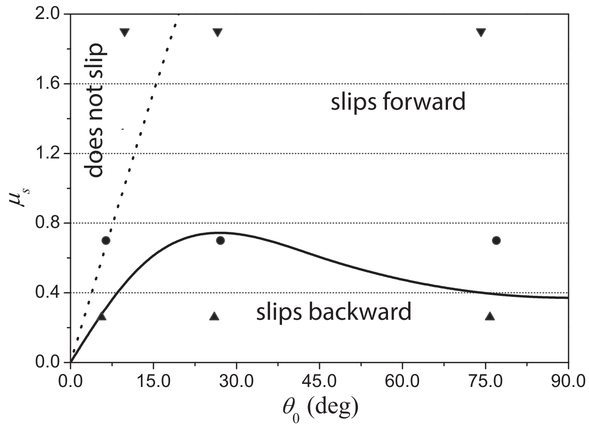

Fig. 7. Static friction coefficients and initial slipping directions. If the value of μ_s falls below the solid line for a given θ_0, the rod will slip backward; if it falls between the solid and dashed lines, the rod will slip forward; if it falls above the dashed line, the rod does not slip at all. The symbols show the slipping directions expected for the angles of release used in the experiments for static friction coefficients of 1.9 for cloth, 0.7 for marble, and 0.26 for steel.

and θ are then determined numerically from α using a finite-difference method (time step of 0.001 s). Next, if the rod is slipping, a_O is computed from Eq. (6) using the values of α, ω, and θ previously calculated; otherwise, a_O is taken to be zero. Finally, the position (x_O) and speed (v_O) of the bottom end of the rod are determined numerically from a_O using the same finite-differences method (time step of 0.001 s). The static friction coefficients were taken from Table I while the kinetic friction coefficients were chosen to optimize the fit to the data. The values used were $\mu_s = 1.6$ and $\mu_k = 0.8$ for cloth, $\mu_s = 0.7$ and $\mu_k = 0.33$ for marble, and $\mu_s = 0.26$ and $\mu_k = 0.185$ for steel. The numerical results are in reasonably good agreement with the experimental data, as shown in Figs. 3, 4, and 6.

Table II shows a comparison between the experimental and theoretical values of θ_s, along with the theoretical value of θ_c, for all of the experiments. In addition, the initial slipping directions observed from experiments and predicted from Fig. 7 are also presented. The agreement between measured and predicted slipping angles θ_s is quite good, as all theoretical values (except one) fall within the margin of error in the experimental data. For the slipping direction, there is a discrepancy between theory and experiment for steel at the smallest initial angle. We believe this discrepancy is due to the difficulty in detecting very small movements of the bottom end of the rod from the videos. In addition, Table II shows that for $\theta_s < \theta_c$, the rod slips forward, while for $\theta_s > \theta_c$ the rod slips backward, as expected from Eq. (2).

An intriguing aspect of this system is that for $\theta_s > \theta_c$, the initial backward slide gives way to a forward slide, whereas for $\theta_s < \theta_c$ the rod only slips forward. This behavior can be qualitatively understood with reference to Eq. (6). For $\theta_s > \theta_c$, the initial backward slide means $a_O < 0$ and hence $\omega^2 \cos\theta < \alpha(\sin\theta - b\mu_K)$. Because this occurs near the beginning of the fall, ω is small and θ is large, making the quantity $\omega^2 \cos\theta$ small. But as the fall progresses, ω increases and θ decreases, both of which act to increase $\omega^2 \cos\theta$. The fact that $\omega^2 \cos\theta$ increases faster than $\alpha(\sin\theta - b\mu_K)$ eventually leads to a positive a_O and a forward slide. For $\theta_s < \theta_c$, the situation is different because the bottom end of the rod remains at rest for a longer period of time. As a result, when sliding occurs ω is already large and θ is small, resulting in a_O being positive for the entire fall.

V. CONCLUSIONS

Although the motion of a uniform thin rod released from rest on a horizontal surface is a well-known system, only theoretical descriptions are available in the literature. In this paper, we have studied the rod′s motion from an experimental point of view using a high-speed video camera. We studied the influence of different types of horizontal surfaces and multiple release angles. For very low release angles the rod does not slip at all, while for higher angles the rod may either slip forward or backward and then forward, depending on the type of horizontal surface. We also presented numerical solutions of the equations of motion that agreed quite well with the experimental data.

It is interesting to note that the present system and methodology can be easily used as a way of determining static frictions coefficients in the classroom. If the rod is released from a near vertical position then the static friction coefficient μ_s can be determined as from a function of the slipping angle θ_s via Eq. (4). As result, students can estimate the static friction coefficient of different surfaces by determining the corresponding slipping angles from videos. Note that, for this purpose, the rod should be released from a near vertical position in order to guarantee that slipping occurs. On the other hand, a natural extension to the present work would be

Table II. The slipping angle θ_s obtained from experiments and numerical simulations. The corresponding values of θ_0 and θ_c, and the initial slipping direction of the rod are also showed. Forward means the bottom end of the rod is moving along the positive x-direction while backward means it is moving along the negative x-direction.

		θ_s (°)			Initial slipping direction	
Surface	θ_0 (deg)	(Experimental)	(Theoretical)	θ_c (deg) (Theoretical)	(Experimental)	(Theoretical)
Cloth $\mu_s = 1.9 \pm 0.3$	74.2	31 ± 1	31.8	39.9	Forward	Forward
	26.6	9 ± 2	7.3	17.4	Forward	Forward
	9.8	—	—	6.5	Does not slip	Does not slip
Marble $\mu_s = 0.9 \pm 0.3$	77.0	35 ± 1	35.9	40.5	Forward	Forward
	27.1	26.5 ± 0.3	27.1	17.7	Backward	Backward
	6.4	—	—	4.3	Does not slip	Does not slip
Metallic $\mu_s = 0.27 \pm 0.1$	75.8	70 ± 1	69.9	40.3	Backward	Backward
	26.0	26 ± 0.3	26.0	17.0	Backward	Backward
	5.6	—	—	4.3	Does not slip	Backward

to investigate the possibly of liftoff during the fall. According to Cross,[5] such a situation should be possible if the rod′s distribution of mass is not uniform, particularly if the rod is top heavy.

Finally, this study demonstrates that high-speed video analysis can be used in the classroom to study situations that occur too quickly to otherwise observe, and to help students relate theory to real-world observations.

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